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Research

Application of the KA-Transformer model to early sepsis prediction: a hybrid network analysis based on time series data

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Abstract

Background Due to the high mortality rate of sepsis, timely identification and intervention in the intensive care unit are essential to improve patient outcomes. Accurate prediction models are critical in addressing the challenge of early sepsis detection.

Methods In this study, we designed a prediction model, KA-Transformer, based on time series data to achieve advance prediction of sepsis. The kernel attention mechanism is introduced in KA-Transformer, which significantly optimizes the problems of limited training samples, many parameters, and uneven sample distribution.

Results By inputting continuous time-series data, the model predicted the area under the wok characteristic curve for subjects with sepsis to be 0.962, 0.944, and 0.984 at the three prediction time points of 1, 6, and 12 h before the onset of sepsis, with accuracies of 92.3%, 93.9%, and 96.1%, respectively.

Conclusions The experimental results show that KA-Transformer outperforms existing methods in terms of accuracy and generalization ability for sepsis prediction, and has the potential to achieve continuous prediction and improve the real-time and reliability of prediction.

Article Highlights

- The KA-Transformer model plays a crucial role in early sepsis detection, risk assessment, and clinical decision-making by enhancing predictive accuracy.
- Deep learning, particularly the KA-Transformer model, outperforms traditional clinical scoring systems in predicting sepsis, offering improved accuracy and adaptability to data dynamics.
- The KA-Transformer model simplifies parameter optimization, captures nonlinear time series features, and addresses sample distribution challenges, achieving high AUC scores in sepsis risk prediction.

Keywords Sepsis · Deep learning · Time series · Kernel attention · Transformer

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1 Introduction

Sepsis is a syndrome of systemic inflammatory response triggered by infection in the body and is accompanied by the potential to cause fatal organ dysfunction [1–3]. Systemic Inflammatory Response Syndrome (SIRS) was first defined as a diagnostic criterion for sepsis in 1991 [4]. With the advancement of medical research, the third international consensus Sepsis-3 [3] published in 2016 updated the definition of sepsis and septic shock and recommended the use of the Sequential Organ Failure Assessment (SOFA) [5] score and the rapid SOFA score (qSOFA) to assess organ dysfunction in patients.

Currently, sepsis poses a serious public health challenge worldwide [6]. According to the World Health Organization [7], there are approximately 30 million cases of infected sepsis worldwide each year, with a mortality rate of 22.5%, which represents 20% of the total global mortality rate. It has been noted that for every hour of delay in the treatment of sepsis, the risk of patient death increases by approximately 4–8% [8, 9]. Therefore, early diagnosis and timely intervention of sepsis are essential to reduce patient mortality.

Physicians commonly use scoring systems such as SOFA and qSOFA to assess the severity of sepsis and predict potential adverse outcomes. However, these scoring systems are not tailored to sepsis patients [10], and these systems may have difficulty performing individualized assessments, especially when the patient's clinical data deviates significantly from the mean, and their predictive accuracy is greatly reduced [11–13].

In recent years, machine learning has made significant progress in the early prediction of sepsis. Machine learning has demonstrated excellent performance in sepsis prediction by training on electronic health record (EHR) data [14–16], with significantly better results than existing clinical scoring systems [17]. Thomas et al. [18] developed the InSight algorithm for classification based on minimal EHR data for early prediction of sepsis in Intensive Care Unit (ICU) patients with an area under the receiver operating characteristic (AUROC) of 0.88, which outperforms traditional clinical early warning systems in terms of classification and calibration. Li [19] et al. introduced the Time-phAsed machine learning model for Sepsis Prediction model (TASP) based on the LightGBM classifier for real-time prediction of sepsis risk, with a utility score of 0.430, sensitivity of 0.859, and specificity of 0.634 on an internal validation set. Shamim [20] et al. proposed the Artificial Intelligence Sepsis Expert (AISE) algorithm based on machine learning using 31,000 ICU patients from two Emory University hospitals as a development cohort and ICU patients from the MIMIC-III database as a validation cohort; AISE showed an AUROC between 0.83 and 0.85 in predicting the 12- to 4-h performance of sepsis with a relatively good prediction performance.

Previous studies have often focused on specific time frames or fixed intervals, inadvertently overlooking the nuanced clinical features and the dynamic progression of sepsis [21]. While the Transformer model offers a solution to account for these dynamics, it faces challenges such as a high parameter count, limited training data, and biased sample distribution. To surmount these obstacles, we introduced the KA-Transformer network in our study,

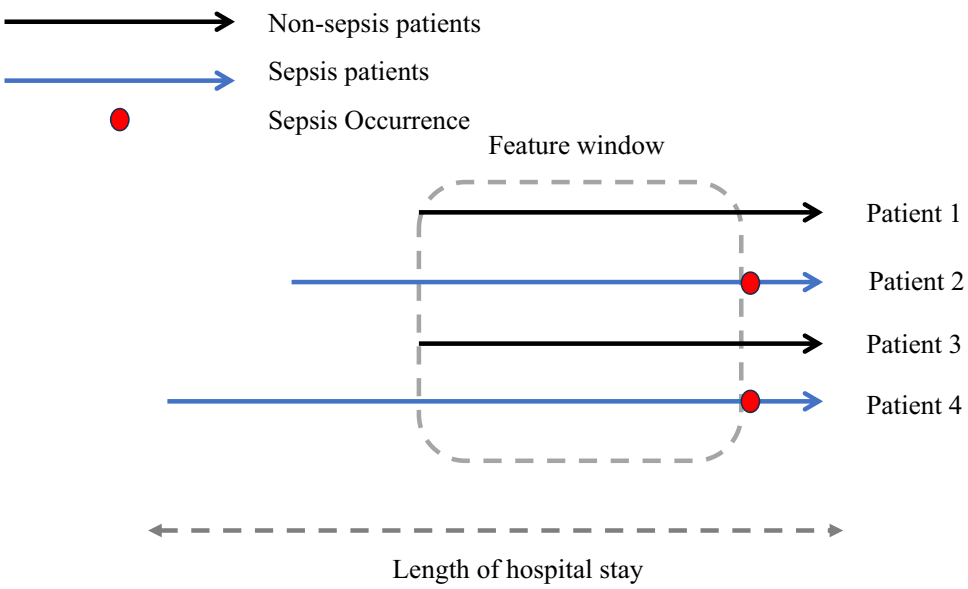


Fig. 1 Schematic diagram of collection feature window

targeting the enhancement of sepsis prediction accuracy within the critical first 1, 6, and 12 h for ICU patients. Building upon the Transformer architecture, our network integrates a novel kernel attention mechanism [22], which supplants the conventional self-attention module. This kernel function, central to our approach, minimizes the parameter load in attention calculations when compared to the dot product method traditionally used. Moreover, these parameters are self-optimized during training, substantially easing the burden of parameter tuning and mitigating the impact of limited data. This innovation not only streamlines the model's adaptability to data fluctuations but also bolsters its capacity to process time series data effectively, capturing the nonlinear aspects intrinsic to sepsis progression.

Section 2 introduces the materials and methods used in the study, including data collection, data preprocessing, the design of the KA-Transformer, as well as the experimental setup and evaluation metrics. It also presents the training process of the KA-Transformer and various performance evaluation methods. Section 4 presents the experimental results and compares the model performance with that of qSOFA, RNN, LSTM, and Transformer. Finally, Section 5 summarizes the main findings of this study and discusses potential directions for future research on the KA-Transformer.

2 Materials and methods

2.1 Data acquisition

We used PhysioNet/Computing in Cardiology Challenge 2019 [23] as training data, which is a database primarily derived from Beth Israel Deaconess Medical Center (BIDMC) and Emory University Hospital (EUH). The dataset exhaustively documents the complete hospitalization cycle of a patient from admission to discharge, covering time-series data for each treatment phase.

In this study, in-depth explorations were conducted in model training and performance optimization, while the MIMIC-IV database [24, 25] (Medical Information Mart for Intensive Care) was used for cross-database validation [26]. The MIMIC-IV database records critical care data from the intensive care units at Beth Israel Deaconess Medical Center between 2008 and 2019. Based on the Sepsis-3 international consensus definition [3], we screened 9,630 adult sepsis patients from the MIMIC-IV database, with positive patient samples representing 24.2% of the total sample size of the database and used these data as an external test set to further assess the generalizability of the model.

The investigator holds a certificate from the National Institutes of Health for the "Protection of Human Research Subjects" training course (No. 49569644) and was authorized to use the MIMIC-IV database through a formal process. In addition, the study was exempt from the requirement for informed patient consent based on ethical review.

2.2 Patient screening and characterization window

To ensure the representativeness of the study population and the reliability of the data, a set of patient inclusion criteria was developed for this study. The patient inclusion criteria were (1) all patients had reached adulthood; (2) all patients needed to have a length of stay of more than 12 h; and (3) for patients with multiple ICU admissions during their hospitalization, only data from their first ICU admission were considered in this study. Exclusion criteria: (1) patients who had been diagnosed with sepsis before entering the ICU; (2) patients who were diagnosed with sepsis after leaving the ICU.

To accurately and promptly identify sepsis, and considering that changes in a patient's condition typically occur gradually, selecting time windows of varying lengths allows for effective capture of the dynamic process from early pathological changes to overt clinical manifestations. Therefore, we used the time of diagnosis as the reference point for sepsis patients and collected continuous time series data at 1-h, 6-h, and 12-h intervals prior to diagnosis in forward increments to construct the predictive model. The starting time was uniformly set as the first recorded data point for each patient, while the ending time was adjusted according to the different prediction feature windows. Within this time frame, we established a uniformly spaced timeline to ensure that the data were aligned at consistent time points, thus ensuring both consistency and comparability of the data. The feature window are illustrated in Fig. 1.

2.3 Inclusion of parameter selection and data preprocessing

In order to ensure the consistency of features between the original dataset and the external dataset, the following 40 clinical indicators, which are closely related to the occurrence of sepsis, were included in this study and used to construct the model:

(1) Vital signs: heart rate, pulse oximetry, temperature, systolic blood pressure, mean arterial pressure, diastolic blood pressure, respiratory rate, end-expiratory carbon dioxide partial pressure.
(2) Laboratory indices: alkali residual, bicarbonate, inhaled oxygen concentration, pH, arterial blood carbon dioxide partial pressure, arterial blood oxygen saturation, aspartate aminotransferase, blood urea nitrogen, alkaline phosphatase, calcium, chloride, creatinine, direct bilirubin, blood glucose, lactate, magnesium, phosphate, potassium, total bilirubin, troponin, erythrocyte pressure volume, hemoglobin, partial thromboplastin time, and white blood cell count, fibrinogen, platelet count.
(3) Clinical Baseline: age, sex, Medical Intensive Care Unit (MICU) management identifier, Surgical Intensive Care Unit (SICU) management identifier, interval between admission and ICU (in hours), and length of stay.

In the data preprocessing stage, this study set a standardized time interval (1 data point collected every hour) to ensure the consistency of the data in the time dimension. For missing values in the dataset, this study used forward padding [27] and backward padding [27] to maintain data continuity. For completely missing features, this study used mean interpolation [28] to ensure data integrity. In view of the category imbalance problem in the dataset, this study applies the Synthetic Minority Over-Sampling Technique (SMOTE) [29] technique on the training set to perform oversampling process to solve the problem of data category imbalance. The specific process is shown in Fig. 2.

2.4 Transformer

Transformer is a deep learning architecture based on a self-attention mechanism for processing sequence data, first proposed by Vaswani [30] et al. in 2017. Compared with traditional recurrent neural networks [31] (RNN) and long-short-term memory networks [32] (LSTM), Transformer does not rely on recursive processing of sequences, but directly computes the interrelationships among elements within a sequence through a self-attentive mechanism, thus capturing global

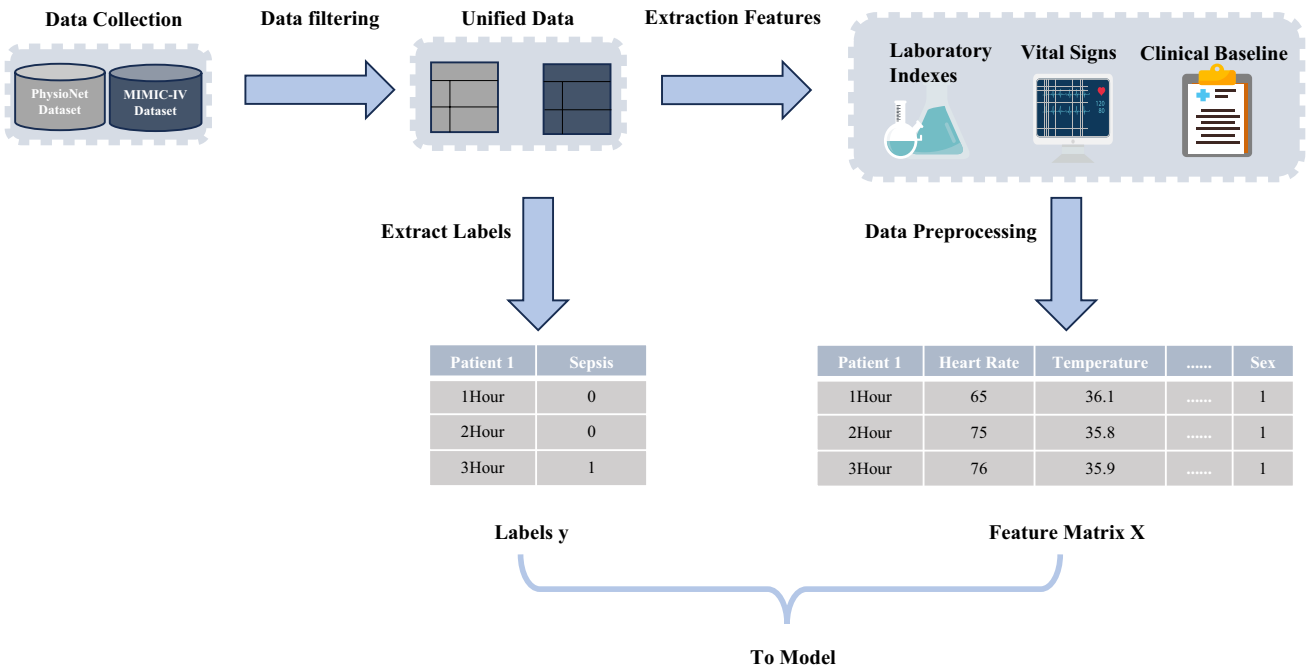


Fig. 2 Workflow chart

dependencies. This mechanism not only enhances the model's ability in dealing with long-term dependency problems, but also significantly improves computational efficiency and parallel processing.

The self-attention mechanism processes the input sequence by generating three vectors, Query (Q), Key (K) and Value (V), and derives the attention score by calculating the dot product between these vectors, which is subsequently normalized using the SoftMax function to obtain the relative importance among the elements. This mechanism effectively captures global dependencies by directly comparing the relationships between the parts of the sequence and greatly improves the ability to process long sequences. The self-attention mechanism is defined as Eq. (1):

$$Attention(\mathrm{Q}, \mathrm{K}, \mathrm{V}) = SoftMax\left(\mathrm{QK}^T / \sqrt{d_k}\right)V \tag{1}$$

where $\mathrm{QK}^T / \sqrt{d_k}$ represents the scaled dot-product attention score, and SoftMax is applied to normalize these scores. The SoftMax function is given by (2):

$$\mathrm{SoftMax}(\mathrm{z})_i = e^{z_i} / \sum_{j=1}^{n} e^{z_j} \tag{2}$$

In this formula, z_i is the raw attention score for the i-th element. The SoftMax function ensures that the normalized attention weights sum to 1, converting the raw attention scores into a probability distribution.

2.5 KA-transformer

The KA-Transformer model proposed in this study is a novel deep learning model designed for the prediction of the onset of sepsis, as shown in Fig. 3. The model includes an input module, a Transformer coding layer, an average pooling layer, and a fully connected layer as a classifier. The model uses kernel attention to process the time series, which enhances the model's ability to deal with nonlinear features by mapping in a high-dimensional space.

The original time series X_{raw} is mapped to a new dimension D through the linear input projection module W_p to generate X_p. Subsequently the location coding incorporates the time point indexes according to the existing methodology into the projected time-series signals. X_p plus the location information is input into L stacked encoder layers, each of which processes the input data through a kernel attention mechanism and a feed-forward network to progressively extract and refine features. The output X_L after processing by all encoder layers is converted into a D-dimensional vector

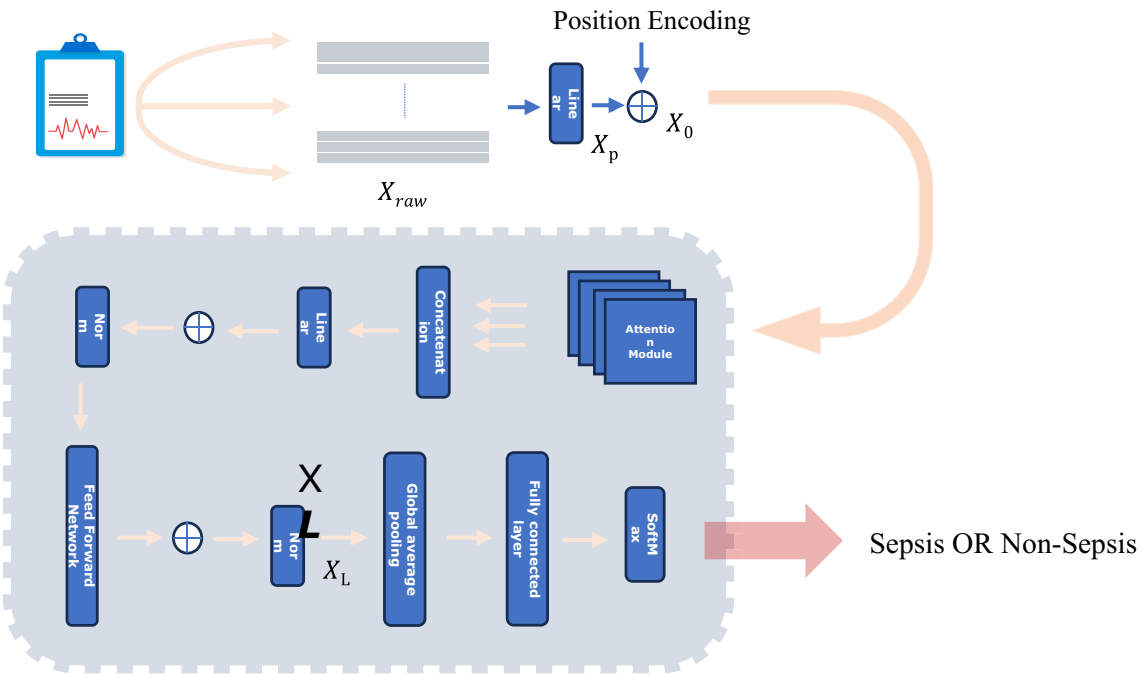


Fig. 3 KA-Transformer architecture diagram

through the global average pooling layer, which is subsequently passed into the fully connected and SoftMax layers for predicting the probability of sepsis at the next time point.

2.5.1 Input modules

The main function of the input module is to transform the input raw time series data X_{raw} from its original feature space to an extended new feature space. This method helps to extend the expressive power of the features, thus enhancing the model's ability to capture the complexity of the data and enabling the model to capture key features in the time series data. In the following Eq. (3), X_{raw} is the raw time series data in 3D, N is the number of features, T is the number of time points, and C is the number of channels.

$$X_{raw} \in R^{N \times T \times C} \tag{3}$$

This module is realized by a predefined linear matrix W_p that maps each data point from the original channel dimension C to a higher dimension D for subsequent encoder processing needs. Where W_p is a transformation matrix from the channel dimension C to the extended feature dimension D, where $D > C$ denotes the extension of the feature dimension. The dimension of the matrix is defined as Eq. (4)

$$W_P \in R^{C \times D} \tag{4}$$

For the input feature vector X_{raw}, by multiplying with the projection matrix $X_p = X_{raw} W_p$, thus mapping the data to a new dimension D. This transformation is done to map the data of each channel to a newly extended D-dimensional feature space. Where the transformed data X_p is adapted to the D-dimensional encoder layer and the data dimensions are given in Eq. (5):

$$X_p \in R^{N \times T \times D} \tag{5}$$

2.5.2 Position encoding

Unlike the traditional Transformer model, the KA-Transformer designed in this study, represents each sample as a $T \times D$ matrix without embedding each sample into a D-dimensional vector. This approach preserves the integrity of the time series for in-depth analysis and reduces the number of projection parameters required to optimize the efficiency and performance of the model. After linear projection, KA-Transformer incorporates time-point information into the projected features through a position encoding module, thus preserving the temporal order of the sequence data.

The position encoding module works in the following way: each time point index is mapped to a D-dimensional vector of real numbers, which are computed by a combination of sine and cosine functions, ensuring the accurate encoding of temporal information:

$$PE(pos, 2i) = sin\left(pos / 10000^{(2i/d_{model})}\right) \tag{6}$$

$$PE(pos, 2i+1) = cos\left(pos / 10000^{(2i/d_{model})}\right) \tag{7}$$

where pos denotes the position index in the time series and d_{model} denotes the dimension of the vector, ranging from 0 to $D/2-1$. This encoded matrix E is added to the processed input X_p to form a new input $X_0 = X_p + E$. This ensures that the model inputs contain not only feature information, but also information about the exact location of the time point.

2.5.3 Kernel attention layer

In order to solve the problems of possible overfitting in the training process due to the insufficient number of samples and the excessive number of parameters in the model, this study introduces the kernel attention mechanism module to replace

the traditional dot product attention mechanism module. The kernel attention mechanism maps the data through a nonlinear kernel function, which effectively captures the complex relationships and dynamic changes in the time series data.

The nonlinear nature and time dependence of the data require the model to capture the complex temporal dynamics and underlying nonlinear relationships. Therefore, this study extracts and processes feature relationships in a high-dimensional space using kernel functions to improve the ability to detect features that are highly relevant to the prediction task, as in Eq. (8):

$$\begin{aligned} A_{i,j} = x_i W_Q, x_j W_K \Rightarrow \\ \phi(x_i), \phi(x_j) = k(x_i, x_j) = K_{i,j} \end{aligned} \tag{8}$$

where x_i and x_j denote different time points in the time series, and $\phi(\cdot)$ is the kernel mapping function that maps the time series to a high-dimensional space to capture the deep temporal dynamics. The similarity between time points is measured by choosing the radial basis function (*RBF*) as the kernel function and calculating $exp(-\beta \parallel xi - xj \parallel 2)$, where β is the learned kernel parameter. This not only reduces the model parameters and decreases the dependence on the number of training samples, but also enhances the model's ability to capture the complexity of the time series by introducing a nonlinear processing mechanism.

2.6 Loss function

In order to effectively address the problem of data category imbalance, this study uses Focal Loss, a loss function designed to deal with the case of extreme category imbalance. Focal Loss reduces the contribution of simple samples to the total loss by modifying the cross-entropy loss so that the model training process pays more attention to samples that are difficult to categorize correctly. The loss function formula is Eq. (9):

$$FL(p_t) = -\alpha_t (1 - p_t)^{\gamma} log(p_t) \tag{9}$$

where p_t denotes the model's probability of predicting the correct category, with a lower value of p_t if the categorization is incorrect, and a higher value if it is not. α_t is a balancing parameter that adjusts the importance of the different categories to cope with the effects of the category imbalance. γ is a positive adjustment factor, which reduces the contribution of the loss of the samples that have been correctly predicted.

2.7 Experimental environment and parameters

Our experimental environment used Intel Core i5-13700 K CPU and NVIDIA GeForce RTX 4060 GPU; the programming language was Python3.12; the deep learning framework was Pytorch2.3, and the dataset was split into a training set (70%) and a test set (30%). During training, the optimizer was Adam, and the learning rate was dynamically adjusted using the cosine annealing algorithm, with the initial learning rate set to 1e-5.The batch size was set to 256, with the model comprising 4 layers and a hidden dimension of 512. A feed-forward network dimension of 2048 was chosen. The number of attention heads was specified as 6, and the kernel size for the kernel attention mechanism was set to 64. To mitigate the risk of overfitting, a dropout rate of 0.2 was applied. The model underwent training for a total of 100 epochs.

2.8 Evaluation indicators

We used a variety of evaluation metrics to measure the performance of the model, including AUROC, sensitivity, specificity, accuracy, precision, recall and F1 score. The specific formulas are as follows:

$$\text{Sensitivity} = \frac{True\ Positives(TP)}{True\ Positives(TP) + False\ Negatives(FN)} \tag{10}$$

$$\text{Specificity} = \frac{True\ Negatives(TN)}{True\ Negatives(TN) + False\ Positives(FP)} \tag{11}$$

$$\text{Accuracy} = \frac{True\ Positives(TP) + True\ Negatives(TN)}{True\ Positives(TP + True\ Negatives(TN) + False\ Positives(FP) + False\ Negatives(FN)} \tag{12}$$

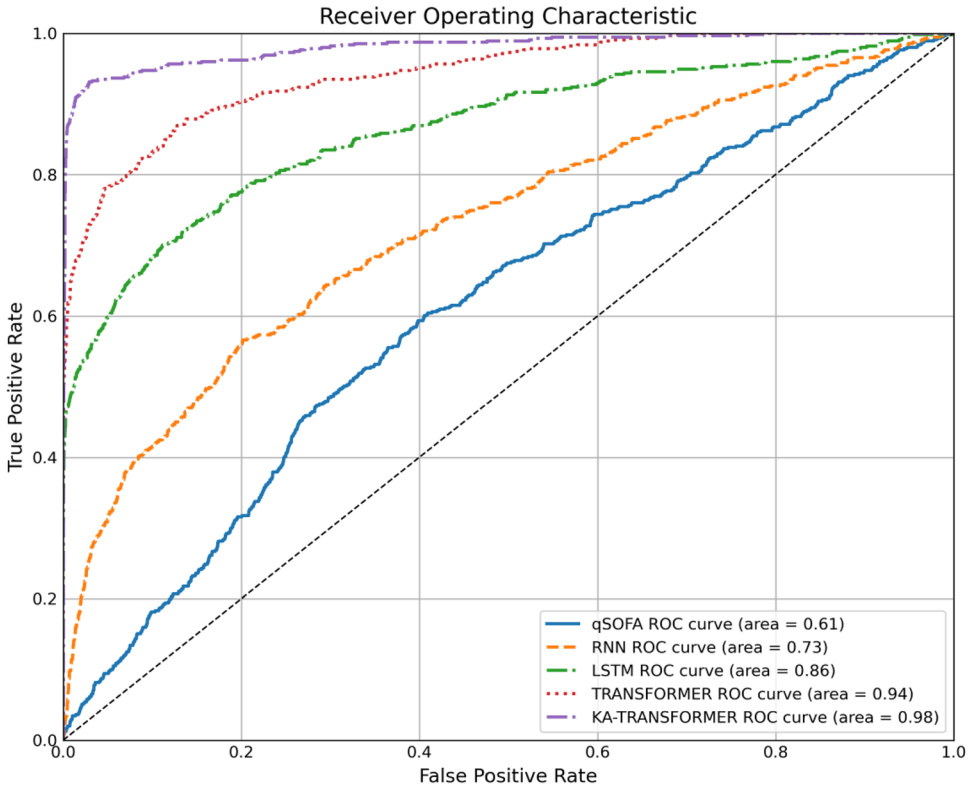


Fig. 4 Feature window is 12-h ROC comparison between KA-Transformer model and other models

Table 1 Time window is 12-h performance comparison between KA-Transformer model and other models

Model	AUC	Acc	Sens	Sepc	F_1
qSOFA	0.610	0.582	0.580	0.575	0.550
RNN	0.728	0.735	0.654	0.721	0.688
LSTM	0.866	0.841	0.721	0.854	0.776
Transformer	0.952	0.924	0.727	0.9065	0.816
KA-Transformer	0.984	0.961	0.829	0.938	0.889

$$\text{Precision} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)+False Positives (FP)}} \tag{13}$$

$$\text{Recall} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)+False Negatives (FN)}} \tag{14}$$

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{15}$$

3 Results

3.1 Model performance

In order to comprehensively evaluate the performance of the KA-Transformer model, this study conducted comparative experiments with commonly used models in the field of time series, including Long Short-Term Memory Networks (LSTMs), Recurrent Neural Networks (RNNs), and Transformer, which are commonly used models. In addition, the clinical scoring method qSOFA was used as a reference standard in this study.

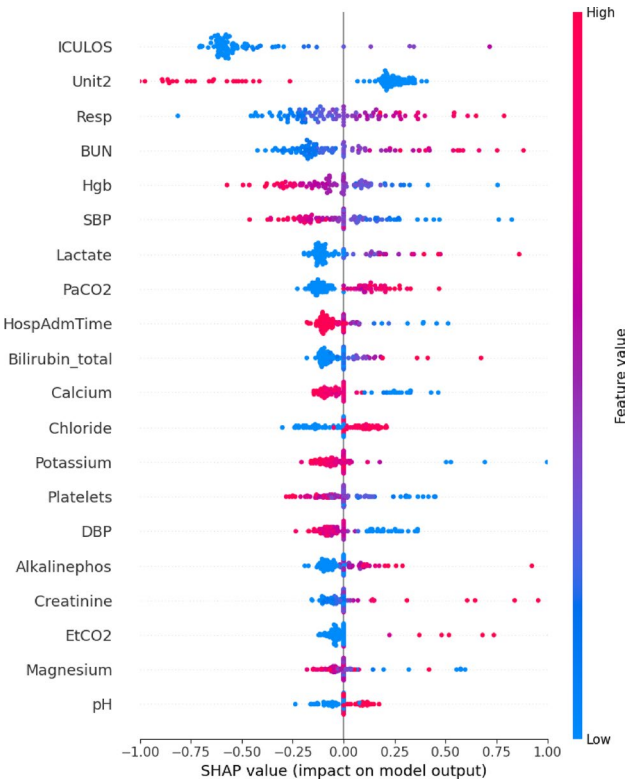


Fig. 5 SHAP values for the KA-Transformer model

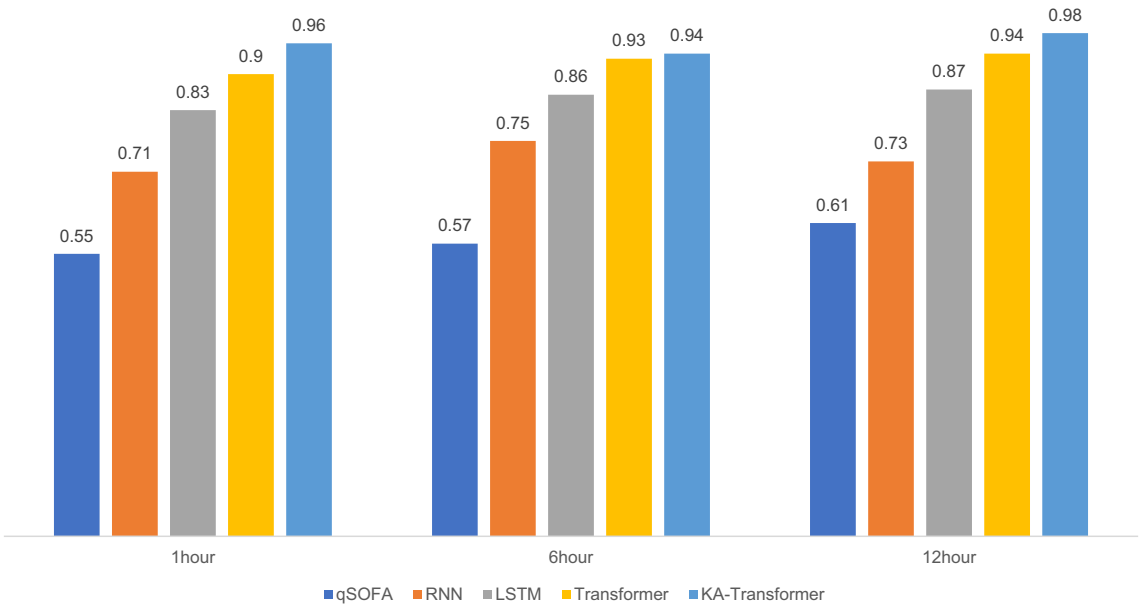


Fig. 6 Prediction results at different time points

During the assessment procedure, a 12-h characterization window was opted for in this research, and the Receiver Operating Characteristic (ROC) curve for the aforementioned model is delineated in Fig. 4. The experimental results indicate that the KA-Transformer model outperforms all other comparative models, achieving an AUROC value of 0.98. As presented in Table 1, a range of statistical metrics was employed to evaluate and compare the performance of various models on the identical test dataset, with the KA-Transformer demonstrating significant enhancements across all performance metrics. Furthermore, the model was weighted utilizing the SHapley Additive exPlanations (SHAP) algorithm to generate visualizations of SHAP values derived from the KA-Transformer model within a 12-h window (Fig. 5).

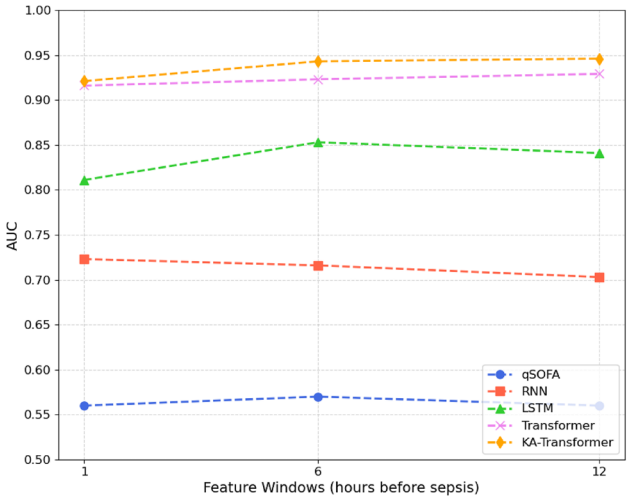


Fig. 7 External validation ROC at different time points

Table 2 Performance comparison between KA-Transformer model and other models in different time Windows

Predicted point in time	Model	AUC	Acc	Sens	Sepc	F_1
1 h	qSOFA	0.56	0.530	0.510	0.550	0.505
	RNN	0.723	0.695	0.640	0.750	0.667
	LSTM	0.811	0.788	0.762	0.814	0.774
	Transformer	0.916	0.889	0.865	0.913	0.885
	KA-Transformer	0.921	0.901	0.889	0.913	0.894
6 h	qSOFA	0.55	0.528	0.505	0.550	0.503
	RNN	0.716	0.698	0.678	0.766	0.695
	LSTM	0.853	0.835	0.820	0.868	0.848
	Transformer	0.923	0.912	0.898	0.937	0.918
	KA-Transformer	0.943	0.933	0.920	0.956	0.941
12 h	qSOFA	0.57	0.547	0.528	0.566	0.520
	RNN	0.703	0.682	0.655	0.709	0.669
	LSTM	0.841	0.821	0.803	0.859	0.826
	Transformer	0.929	0.915	0.910	0.934	0.920
	KA-Transformer	0.946	0.936	0.924	0.968	0.943

In this study, we compared the prediction performance of different network models by adjusting different feature windows and collecting patient time-series data 1 h and 6 h prior to sepsis diagnosis. The KA-Transformer model had the best prediction performance in the 12-h window, and the performance declined as the feature window was narrowed (Fig. 6).

3.2 External database validation

In order to verify the generalizability of the KA-Transformer model, this study uses data from MIMIC-IV for external validation. The prediction results for different feature windows are shown in Fig. 7. The AUC of the KA-Transformer model reaches 0.94 when predicting 12 h in advance, which is the best performance; when predicting 6 h and 1 h in advance, the AUC of the model is 0.94 and 0.92, respectively, which is always better than other models.

In addition, Table 2 shows a variety of statistical metrics for each model on the external validation dataset under different feature windows. It can be seen that the KA-Transformer model performs well in all the metrics under each feature window, especially under the 12-h window, where the accuracy and F1 score reach 93% and 90%, respectively.

4 Discussion

In this study, we designed a Transformer framework model, KA-Transformer, for early prediction of sepsis in ICU patients. Using the PhysioNet/Computing in Cardiology Challenge 2019 dataset, the inputting continuous time-series data of patients 1 h, 6 h and 12 h before diagnosis to different network models. Our proposed KA-Transformer achieved the best performance at all three time points, demonstrating the effectiveness of the KA-Transformer model in early identification of sepsis in ICU patients.

The emergence of Transformer [30] has provided new ideas for constructing sepsis prediction models. The self-attention mechanism in Transformer performs well in the task of sepsis prediction with respect to long-range dependencies in sequential data. Wang [33] et al. proposed a multimodal Transformer model that enabled the prediction of sepsis within 36 h of ICU admission by combining physiologic time series data with clinical notes. Tang [34] et al. utilized CNN-Transformer and LSTM-Transformer for advance prediction of sepsis, and the model exhibited about 20% performance improvement compared to traditional recurrent neural network, with an accuracy of 0.964, a precision of 0.956, a recall of 0.967, and an F1 score of 0.959. In contrast, the KA-Transformer proposed in this study showed significant performance improvement over the traditional Transformer and recurrent neural network models, especially in predicting the incidence of sepsis 12 h in advance, which demonstrated the potential of the KA-Transformer in the field of clinical prediction and highlighted the importance of choosing the appropriate feature window on the effectiveness of the model. It also outperformed the clinical evaluation metric qSOFA by a considerable margin. qSOFA relies on three static clinical indicators, which cannot effectively process time-series data or capture the dynamic progression of a patient's condition. The qSOFA score's limitations are further compounded by its inability to fully incorporate a patient's comprehensive medical history or utilize real-time monitoring data, which are critical for early and accurate sepsis detection. This exploratory study not only opens a new direction in medical research, but also provides a valuable reference for future healthcare practice.

We employed the SHAP methodology to enhance the model's interpretability, thereby empowering physicians to pinpoint key sepsis risk factors with greater precision. The experimental findings, as depicted in Fig. 6, underscored that ICU length of stay, SICU management identifiers, hemoglobin levels, and BUN are pivotal in the genesis of sepsis. An extended ICU stay was found to be positively correlated with sepsis incidence, potentially attributable to increased vulnerability to hospital-acquired infections. Patients in the SICU, often presenting with more severe conditions, face an elevated risk of postoperative wound infections following intricate surgical procedures, substantially amplifying sepsis likelihood.Moreover, hemoglobin levels, a vital gauge of blood oxygenation and overall health, emerged as a significant contributor. Reduced hemoglobin levels can precipitate malnutrition, chronic illnesses, and blood loss, thereby augmenting the propensity for infections and, by extension, sepsis risk. Additionally, heightened BUN levels were strongly associated with compromised kidney function or internal milieu disruptions, which can diminish immune function and heighten the susceptibility to septic infections.Leveraging SHAP, we have successfully elucidated these critical variables and their nexus to sepsis, equipping clinicians with a deeper comprehension of the associated risk factors. This enhanced understanding is instrumental in guiding more informed and effective clinical decision-making processes.

In this study, the model was externally validated using the MIMIC-IV dataset. The experimental results show a decrease in the predictive performance of the model compared to the internal dataset, which is mainly due to the differences in statistical characteristics and clinical practices between the different datasets. Despite the decrease in prediction performance, the KA-Transformer model still significantly outperforms the other comparative models at the three prediction time points, demonstrating its excellent generalization ability and superior overall classification performance.

There are still some limitations to this study: for the problem of adaptation to geographic and health system differences, future research needs to further explore how to improve the validity and application value of the model

in a wider range of data environments through adaptation and optimization; at the same time, there are usually a large number of missing values in the EHR data, which may lead to some estimation bias in the model predictions.

5 Conclusion

In this study, we designed a sepsis risk prediction model based on the Transformer architecture, which introduced the kernel attention mechanism, significantly simplified the number of parameters required for model training, and effectively solved the problem of insufficient samples. In addition, the kernel attention can efficiently capture the nonlinear features in the time series data, which enhances the ability to adapt to the dynamic changes of the data. The input module of the model maps the original time series into a high-dimensional space, preserving its local features. The location coding module integrates the time point information into the features, which maintains the temporal sequence of the time series and improves the processing efficiency and performance of the model.

To address the challenge of uneven sample distribution, this study introduces a focal loss function (FLF) to focus more intensively on samples that are difficult to classify, thereby significantly improving the prediction accuracy of the model. In tests at multiple prediction time points, the KA-Transformer model consistently outperformed other models. Under the 12-h window, the model achieved an optimal AUC of 0.94, further validating its superior predictive performance in the field of sepsis risk prediction.

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Availability of data and materials The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate The study utilized de-identified data from the MIMIC-IV database, and as such, was deemed exempt from informed consent by the ethical review board, aligning with the Declaration of Helsinki.

Consent for publication None.

Competing interests The authors declare no competing interests.

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